

Review of confidence intervals and introduction to the bootstrap

Overview

Review and continuation of sampling distributions and confidence intervals

If there is time:

- The bootstrap

Announcement

Homework 5 has been posted!

It is due on Gradescope on **Sunday October 11th at 11pm**

- **Be sure to mark each question on Gradescope!**

The material this week is going to be a bit more conceptually challenging

Please attend the practice sessions and office hours to reinforce your understanding!

Review of confidence intervals

Question: What is a confidence interval?

A **confidence interval** is an interval computed by a method that will contain the *parameter* a specified percent of times

- i.e., if the estimation were repeated many times, the interval will have the parameter x% of the time

The **confidence level** is the percent of all intervals that contain the parameter

Think ring toss...

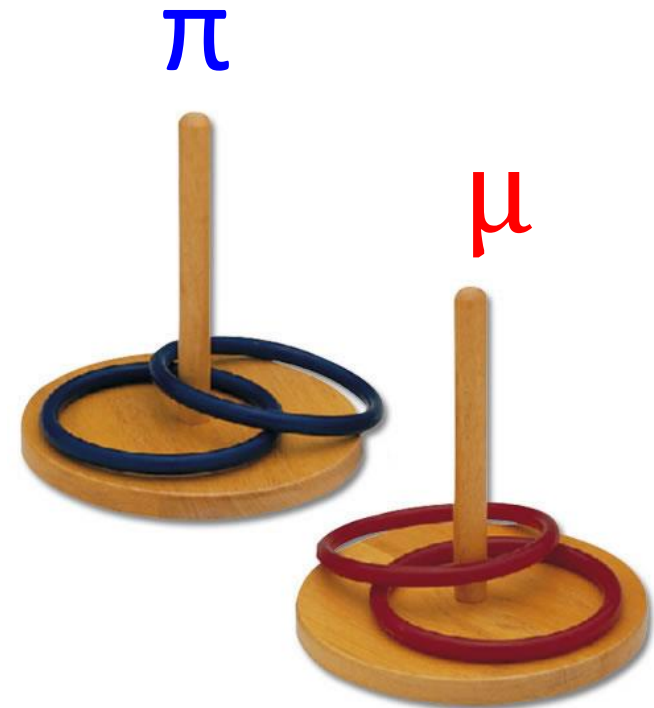
A parameters exists in the world

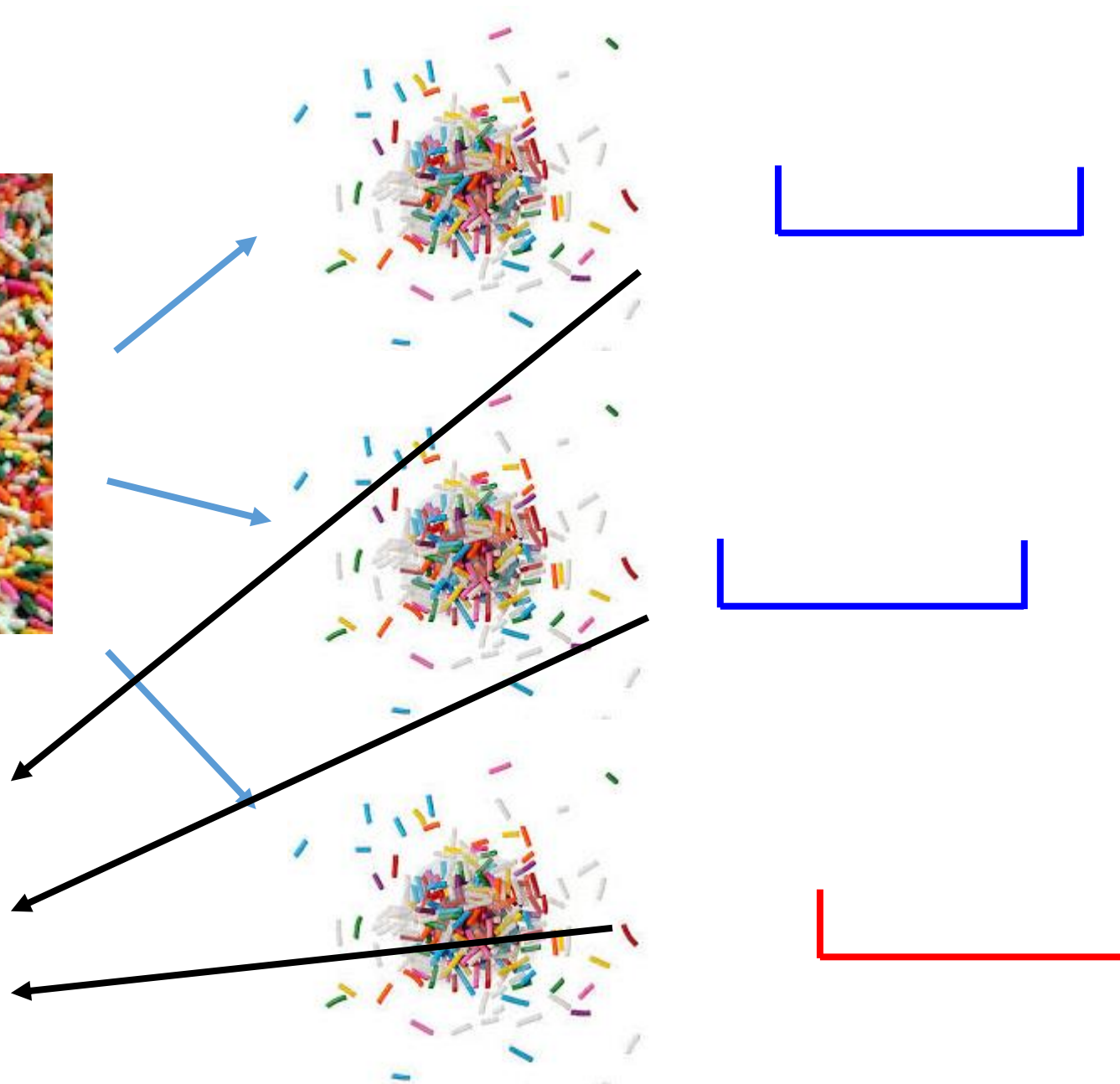
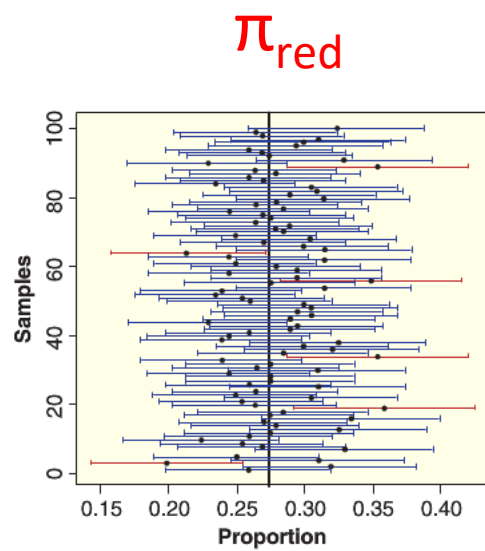
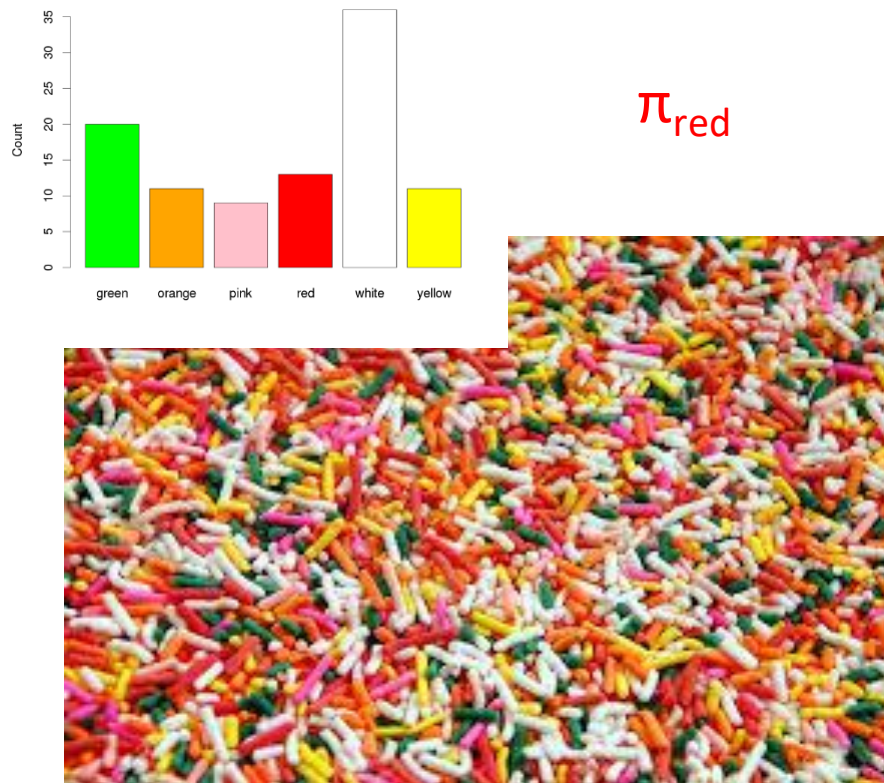
We toss intervals at them

- These are our confidence intervals

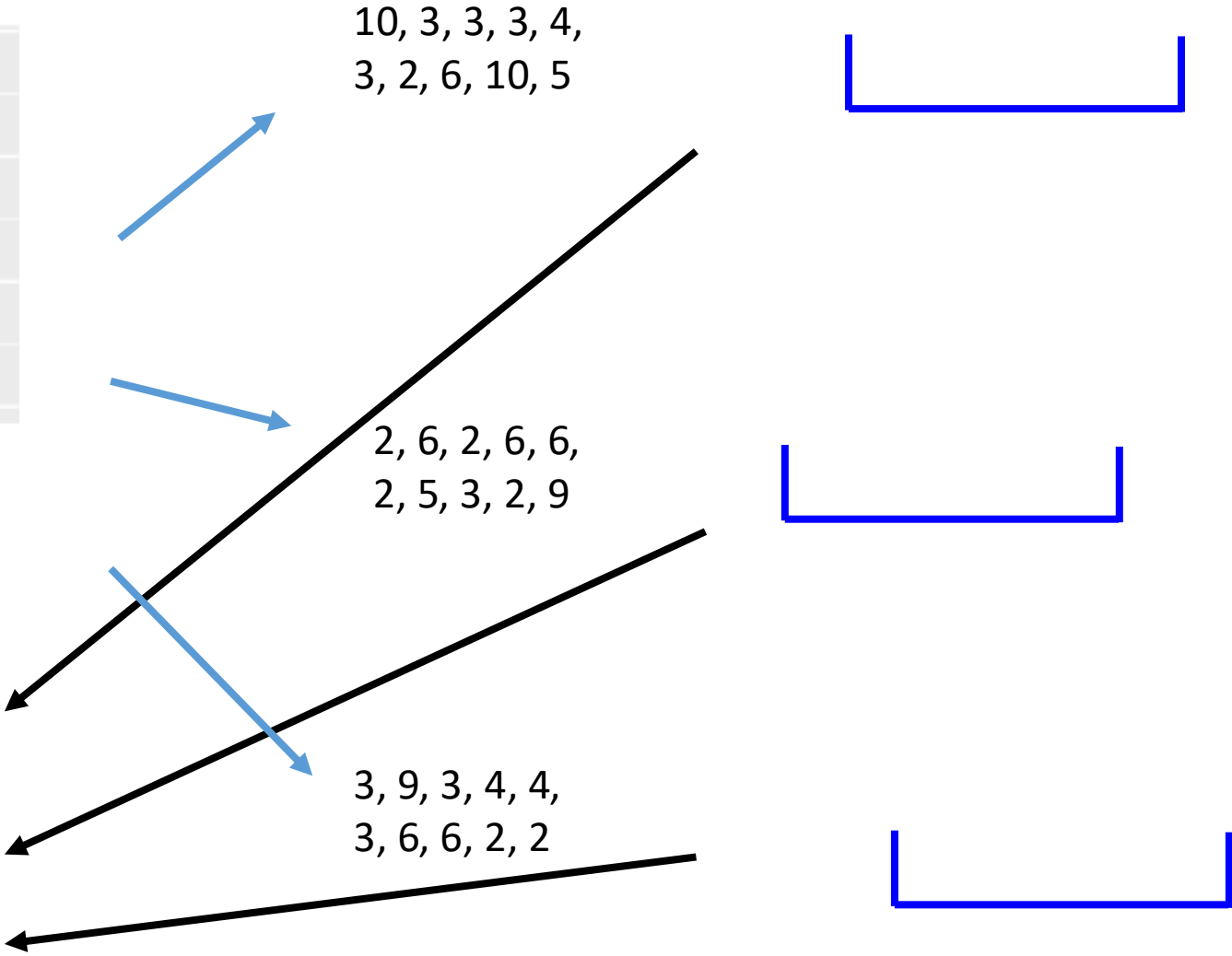
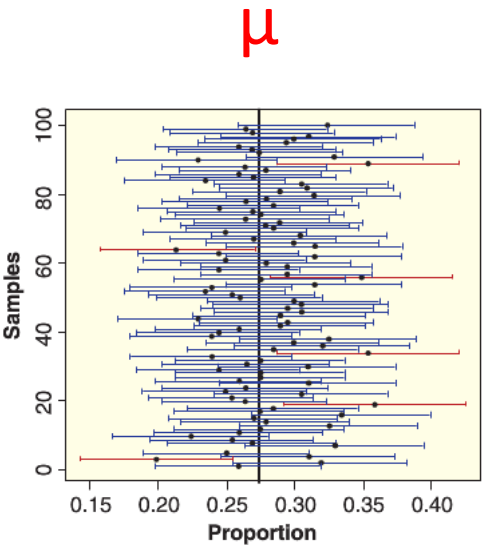
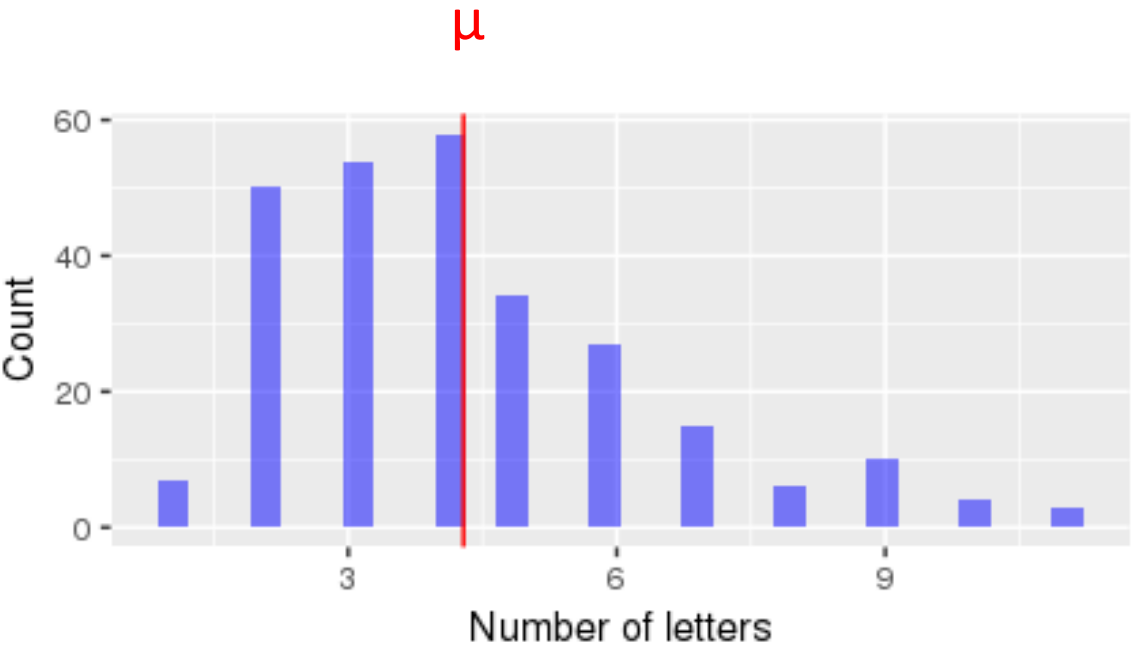
95% of those intervals capture the parameter

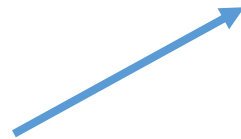
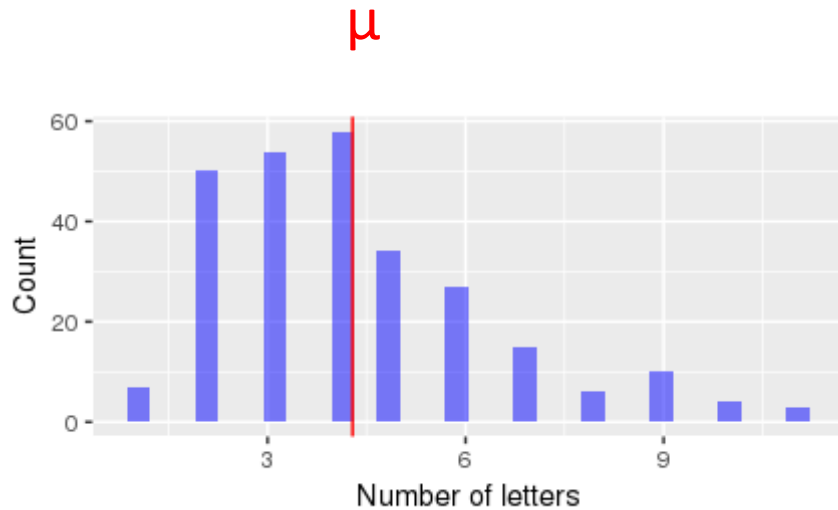
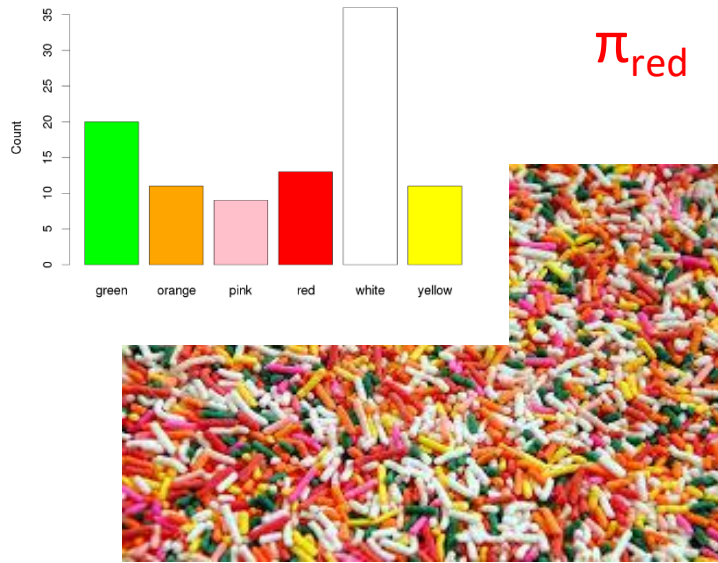
- (95% is our confidence level)



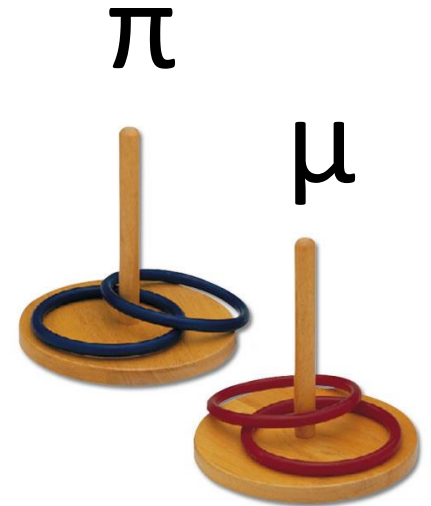


Gettysburg address word length sampling distribution





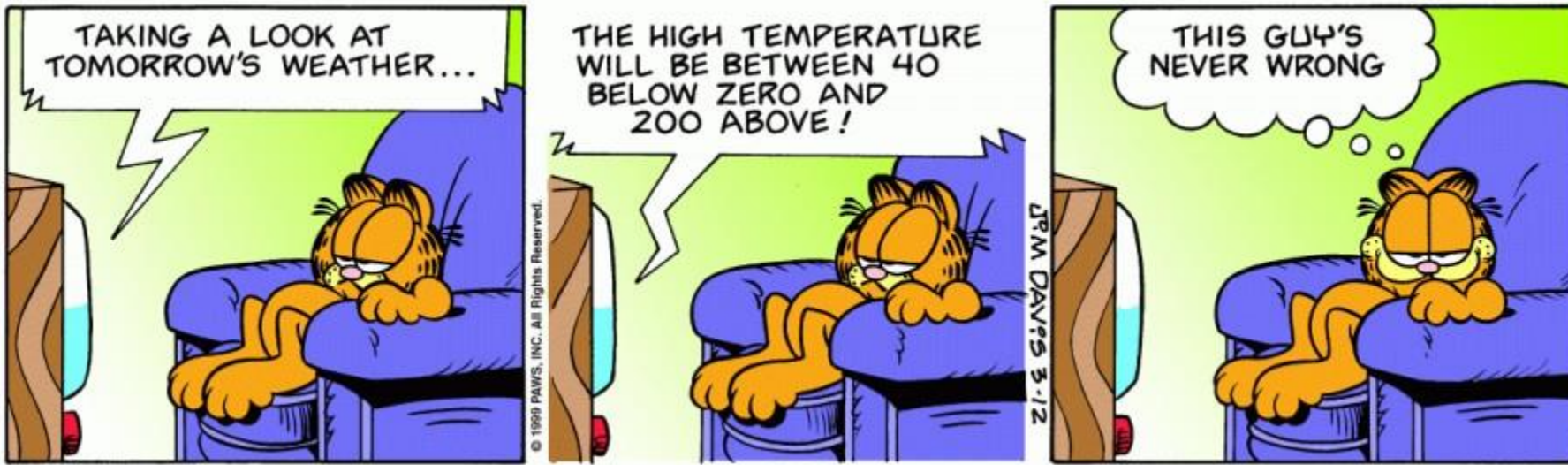
10, 3, 3, 3, 4,
3, 2, 6, 10, 5



For a 95% confidence level, 95% of the intervals we create will have the parameter in them

Confidence Intervals

There is a tradeoff between the **confidence level** (percent of times we capture the parameter) and the **confidence interval size**!



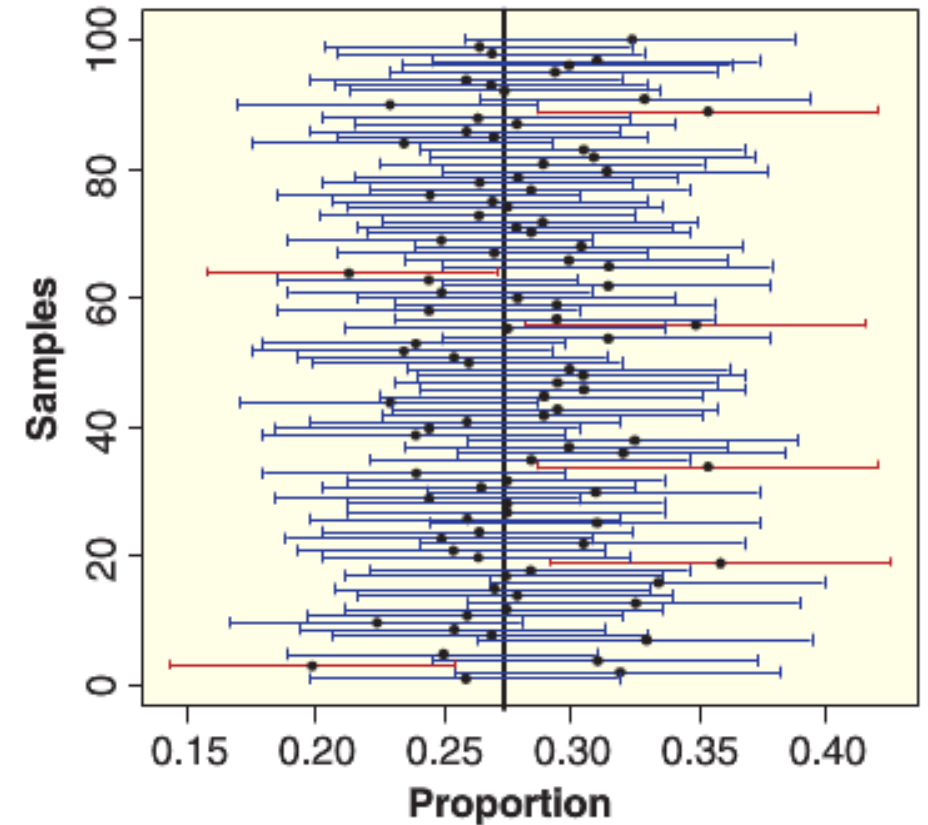
Confidence Intervals

Q: For any given confidence interval, do we know whether it has really captured the parameter?

- No ☹️

But we do know that if we create 100 intervals, ~95 of these intervals will have the parameter in it

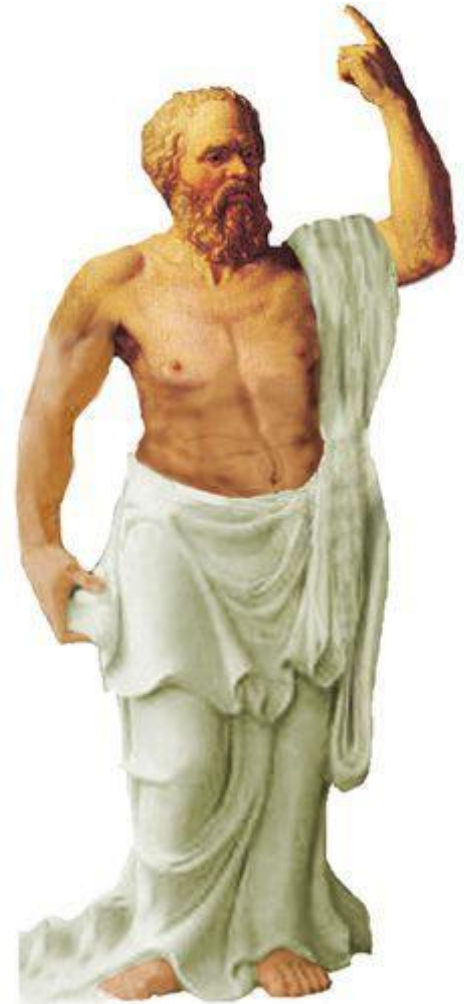
(for a 95% confidence interval)



Review of sampling distributions

Review of confidence intervals and sampling distributions

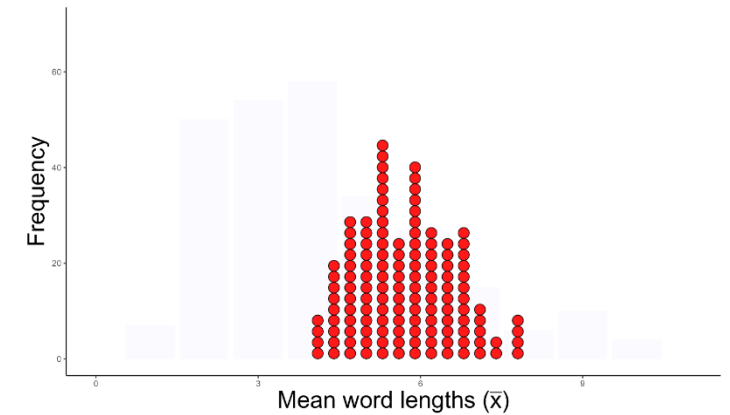
Question₀: Who is this?



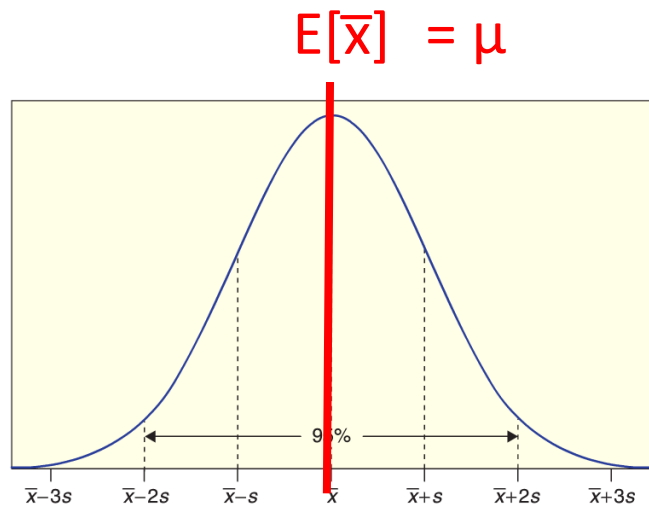
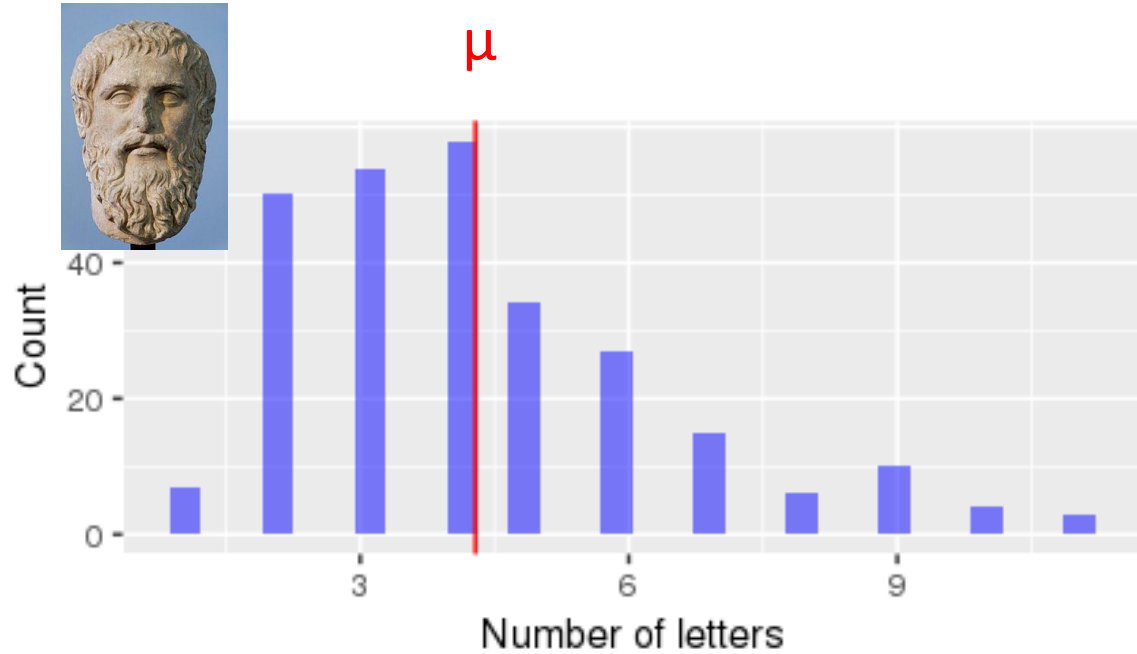
Sampling distributions

Q₂: What is a sampling distribution?

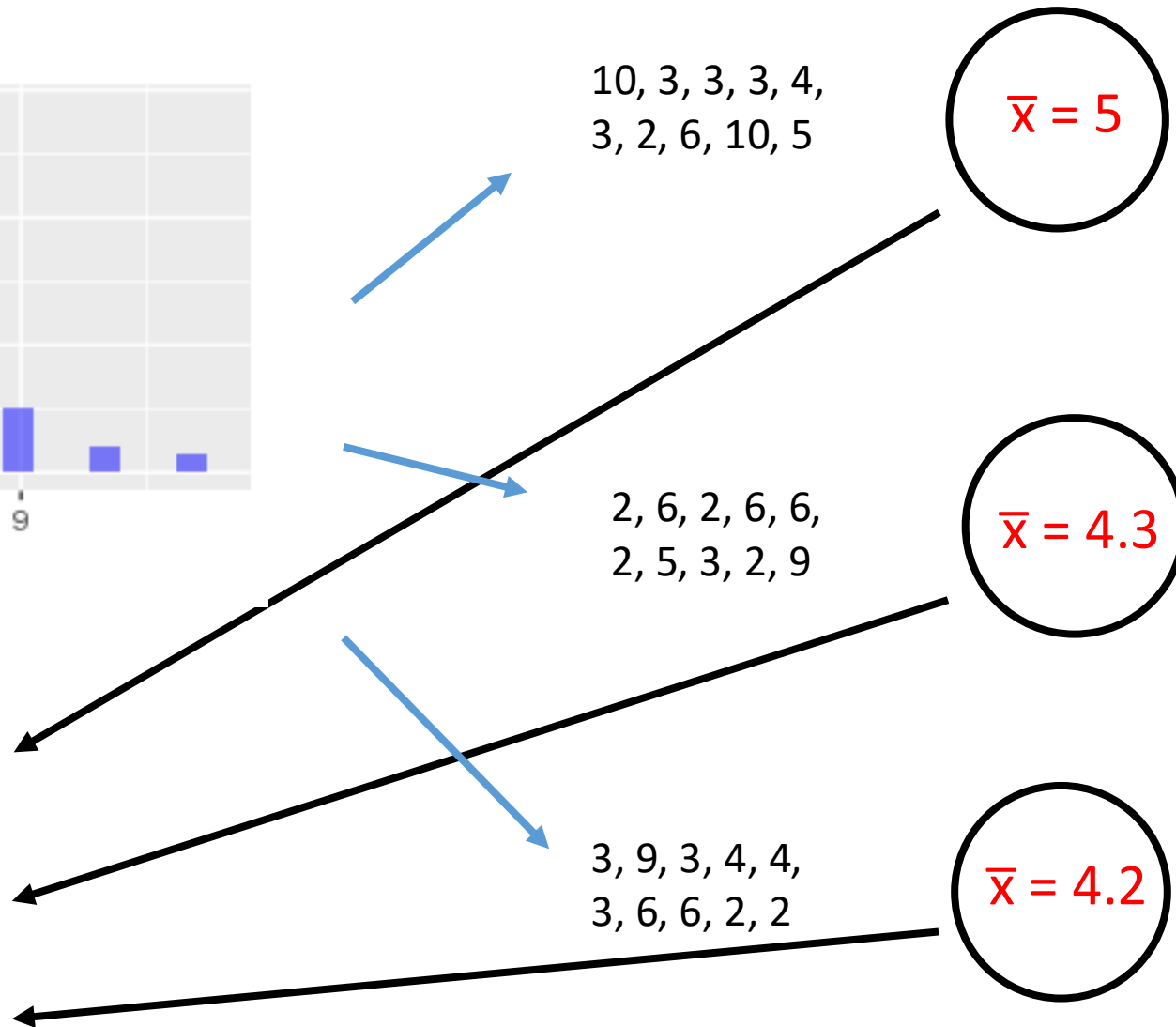
Q₃: What does a sampling distribution show us?



Gettysburg address word length sampling distribution



Sampling distribution!

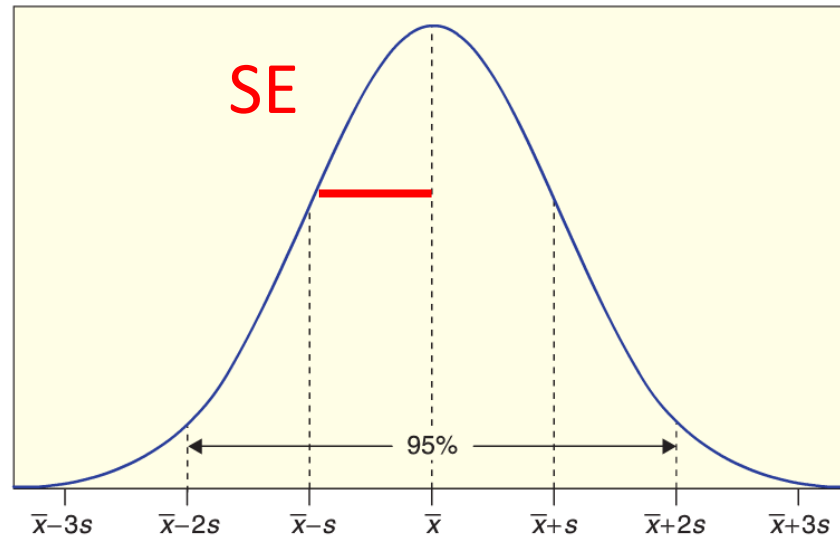


Unbiased: $E[\bar{x}] = \mu$

The standard error

Q₄: What is the **standard error**?

Q₅: What symbol do we use to denote the standard error?



Sampling distribution in R

Q₆: Suppose we have a function called `get_sample()` that can generate samples from a population

How could we estimate the SE of the mean using R?

```
sampling_dist <- do_it(10000) * {  
  curr_sample <- get_sample()  
  mean(curr_sample)  
}
```

What symbol should
we use for this quantity?



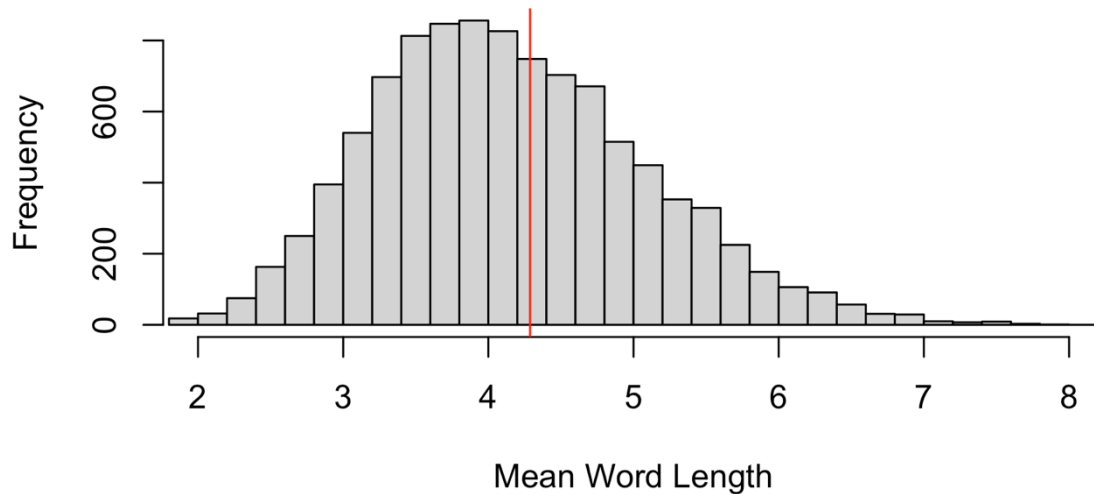
```
SE_mean <- sd(sampling_dist)
```

Q₇: What are two ways that sampling distribution changes with a larger sample size n ?

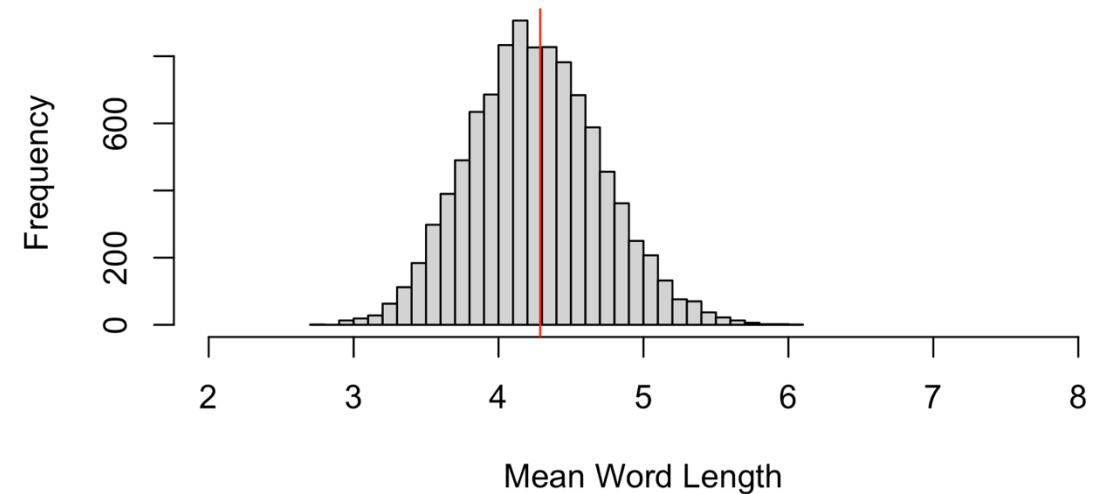
As the sample size n increases:

1. The sampling distribution becomes more like a normal distribution
2. The sampling distribution statistics become more concentrated around population parameter
 - i.e., the SE becomes smaller

Sampling Distribution with $n = 5$



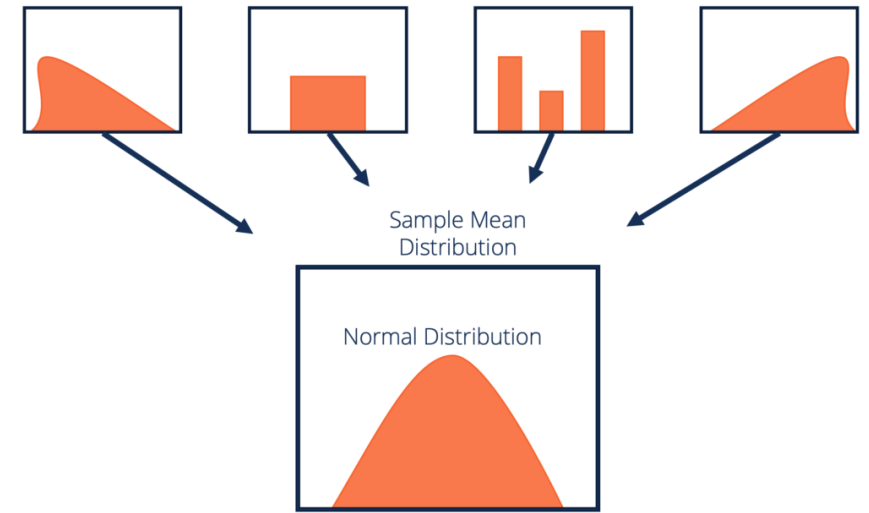
Sampling Distribution with $n = 20$



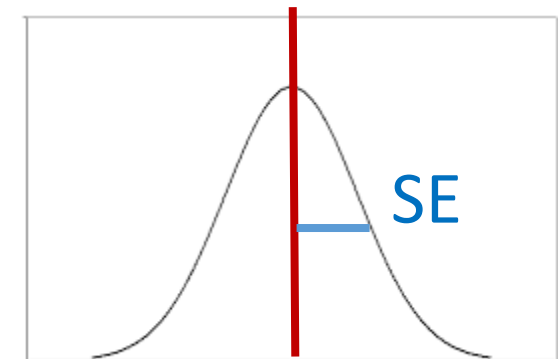
Central limit theorem

For random samples with a sufficiently large sample size (n), the distribution of sample statistics for a **mean** ($\bar{x}$) or a **proportion** ($\hat{p}$) is:

- Normally distributed
- Centered at the value of the population parameter



Parameter



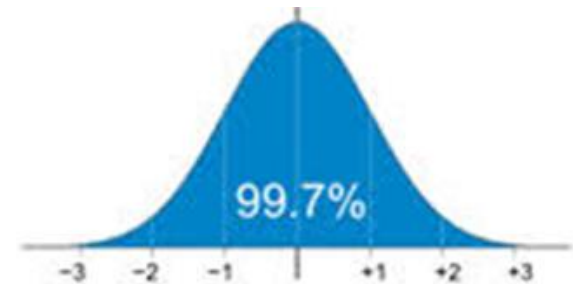
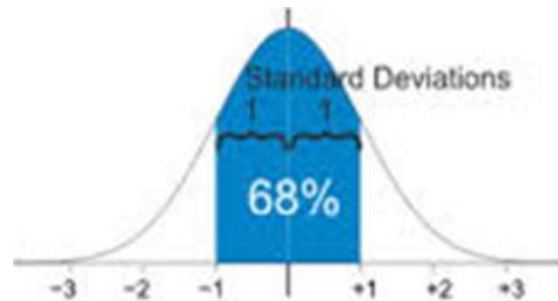
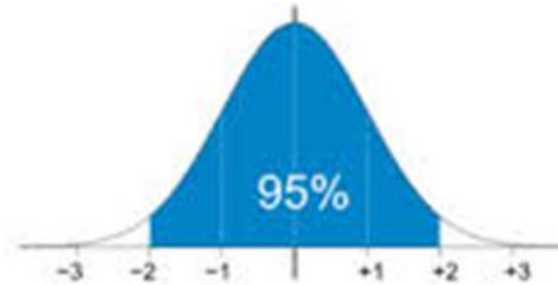
Shapes of sampling distributions

Q₈: What is a commonly seen shape for sampling distributions?



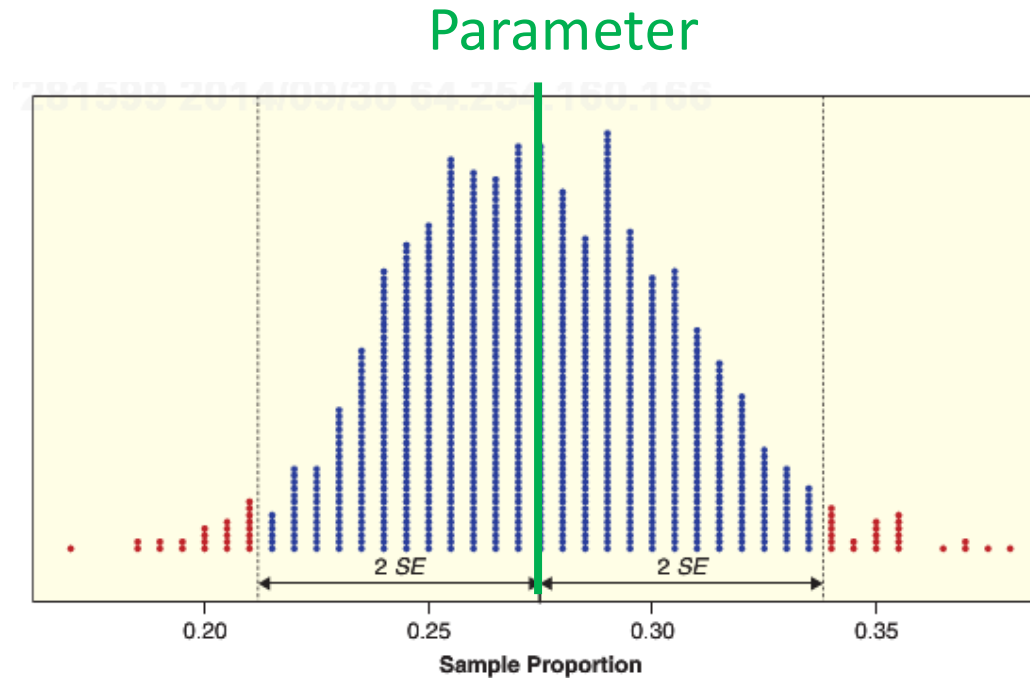
Normal distributions

Q₉: For a normal distribution, what percentage of points lie within 2 standard deviations for the population mean?



Sampling distributions

Q₁₀: For a sampling distribution that is a normal distribution, what percentage of *statistics* lie within 2 standard deviations (SE) of the parameter value?



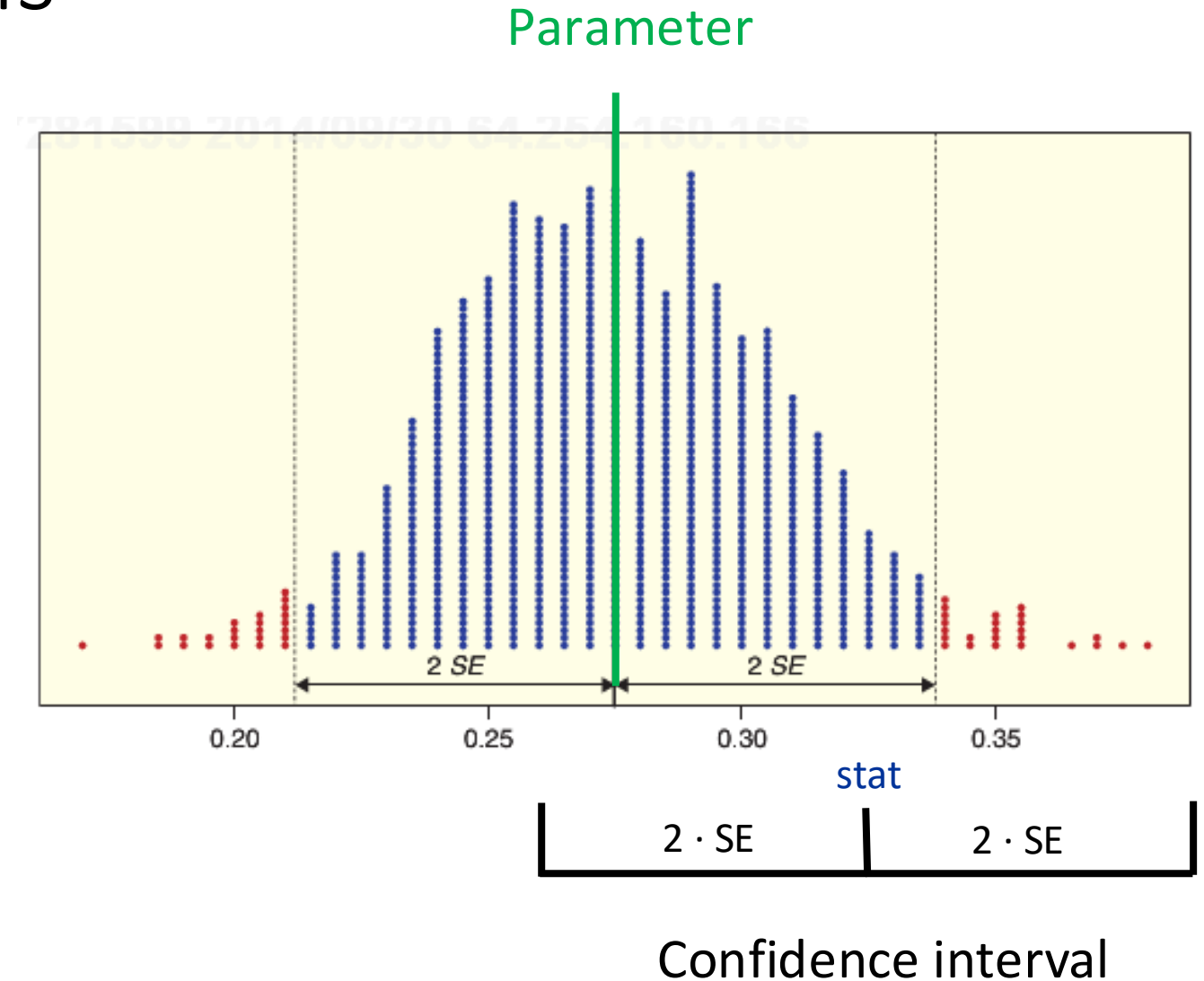
Q₁₁: If we had a *statistic value* and the value of the *SE*, could we compute a 95% confidence interval?

Confidence intervals

Q_{11} : Suppose we are going to randomly chosen statistic value

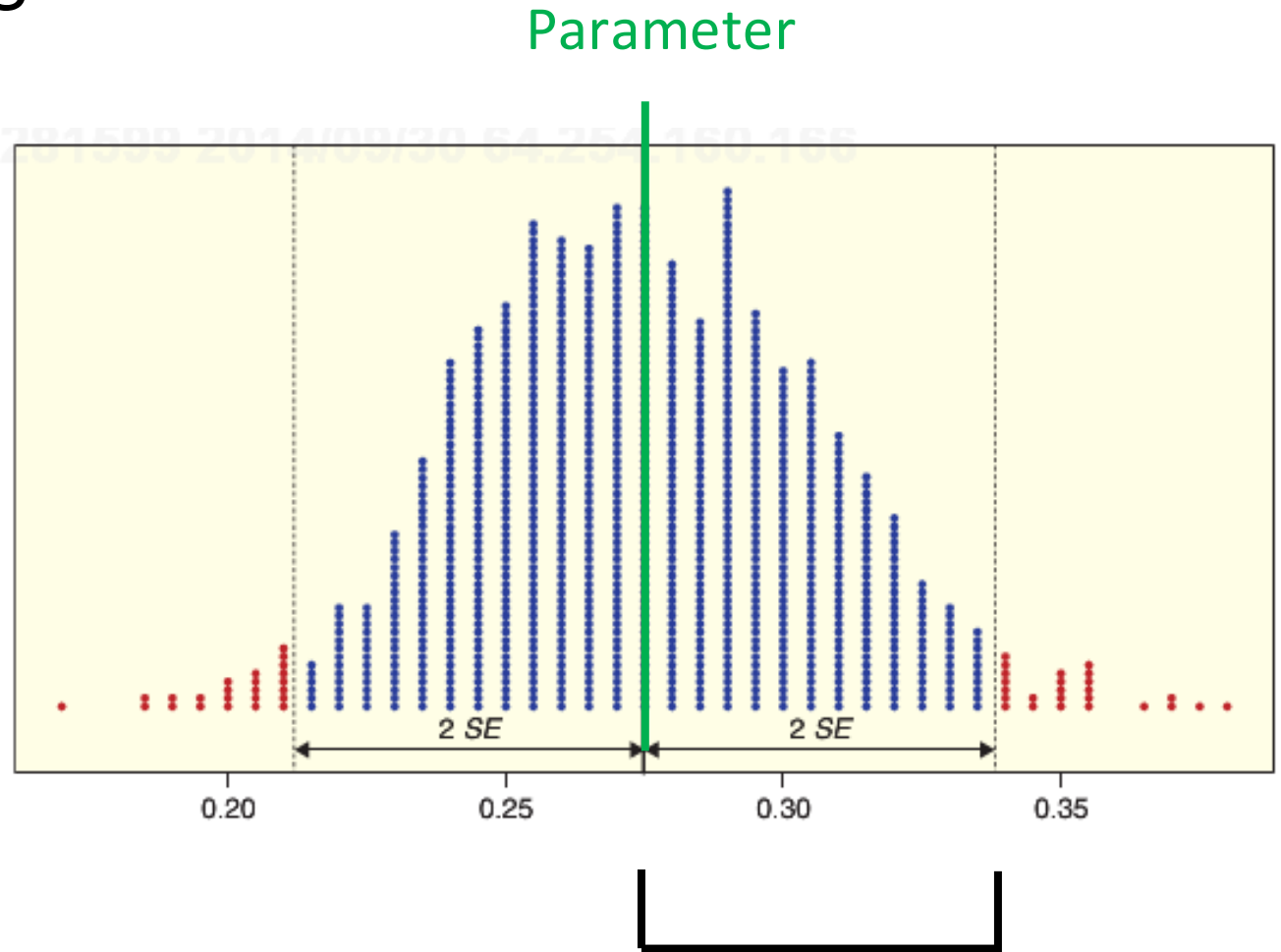
And we are going create an interval centered at the statistic value

How large would the interval need to be to overlap with the parameter 95% of the time?



Confidence intervals

Q₁₂: What is a formula can we use to calculate 95% confidence intervals?

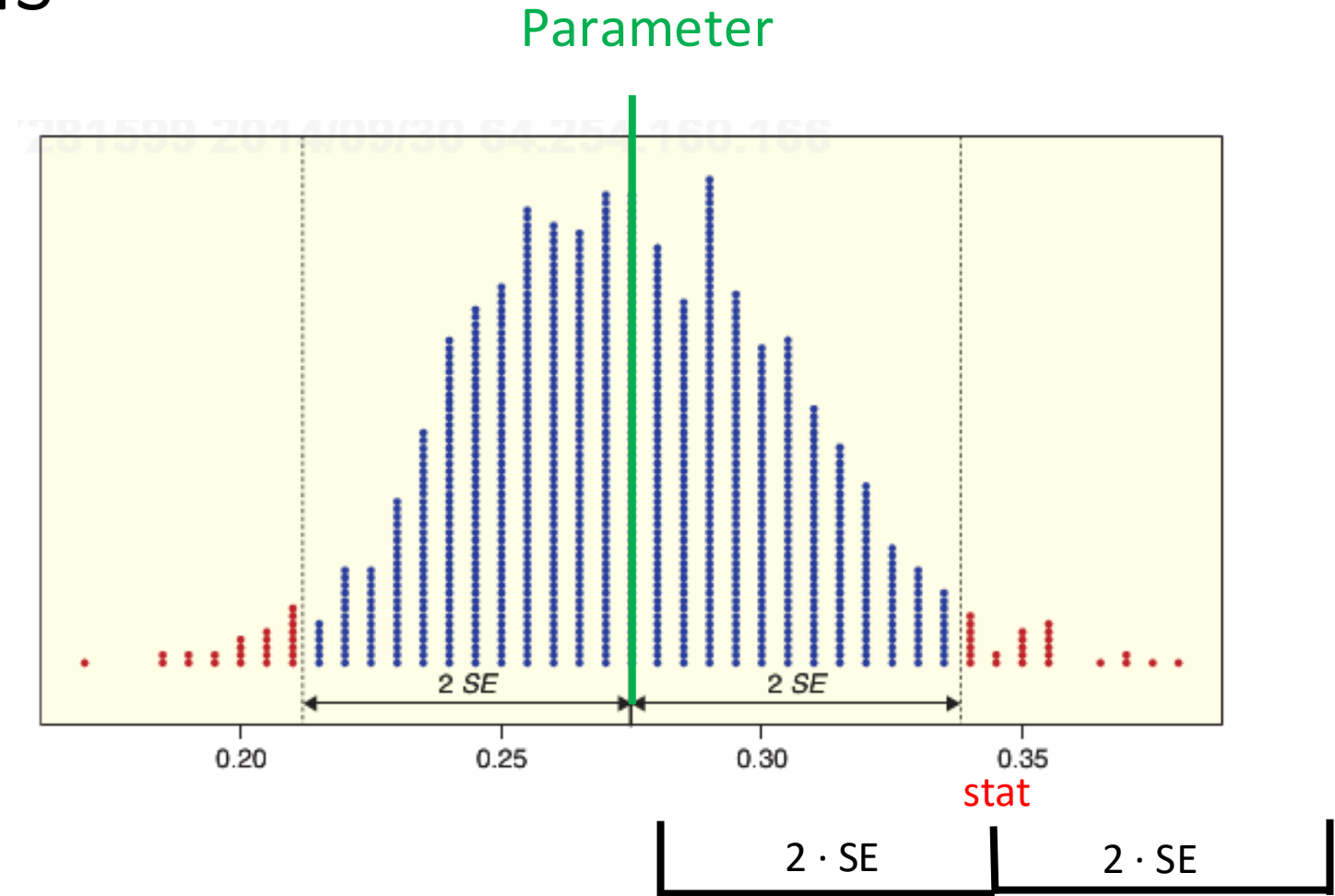


95% confidence interval: stat $\pm 2 \cdot SE$

Q₁₃: What is this quantity called?

Confidence intervals

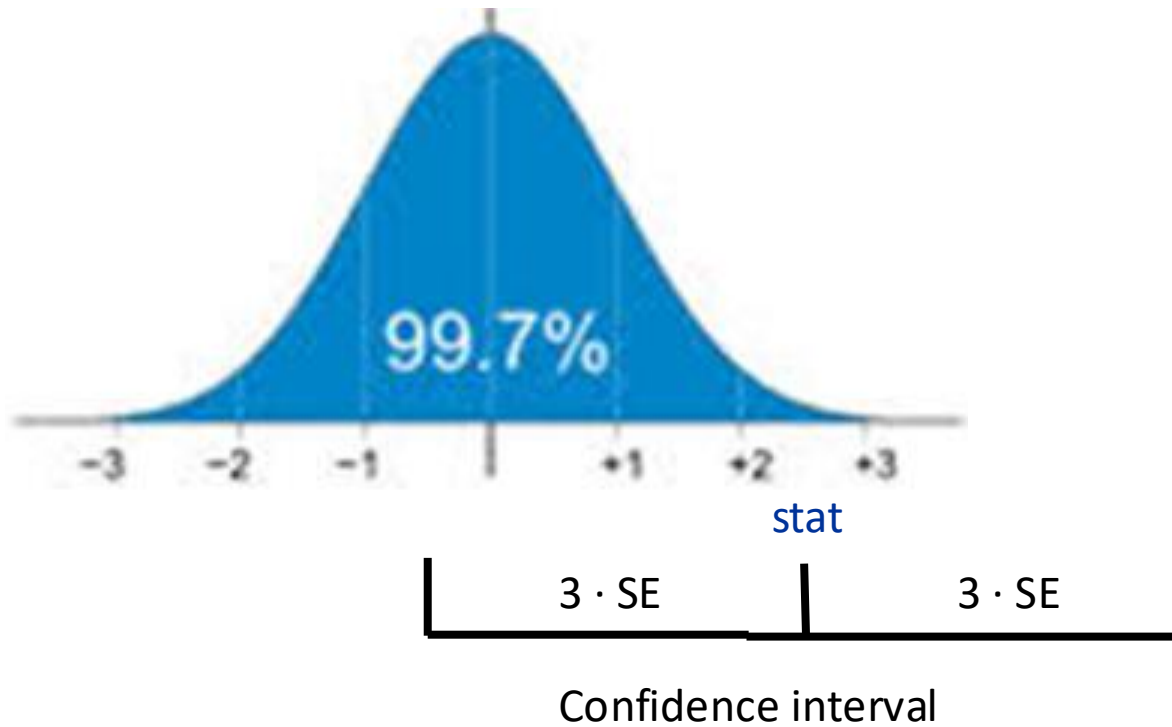
Q₁₄: How frequently do
95% confidence intervals
fail to capture the
parameter of interest?



95% confidence interval: $\text{stat} \pm 2 \cdot \text{SE}$

Confidence intervals for other confidence levels

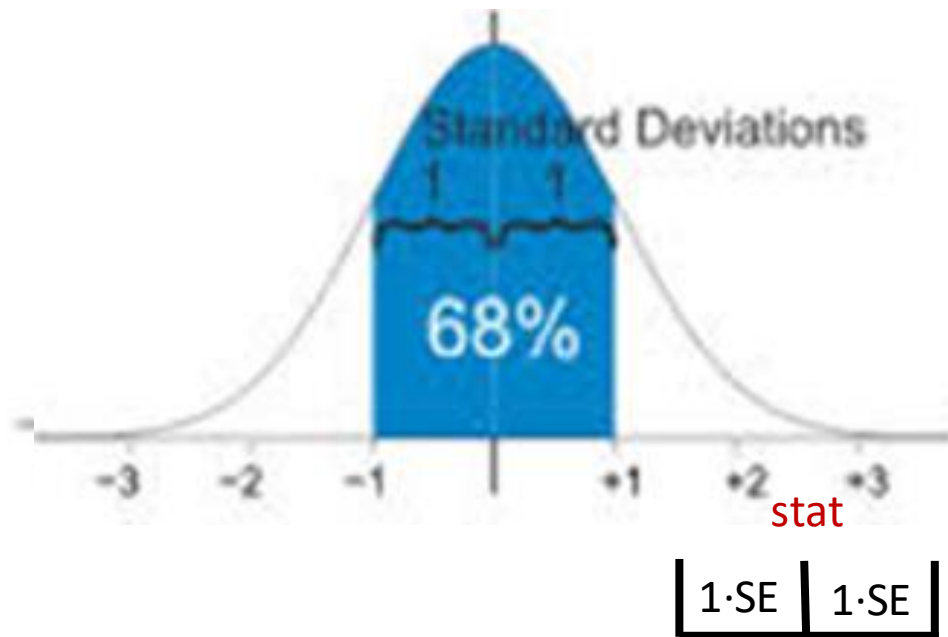
Q₁₅: How could we get a 99.7% confidence interval confidence level?



$$CI_{99.7} = \text{stat} \pm 3 \cdot SE$$

Confidence intervals for other confidence levels

Q₁₆: How could we get a 68% confidence interval confidence level?



Confidence interval

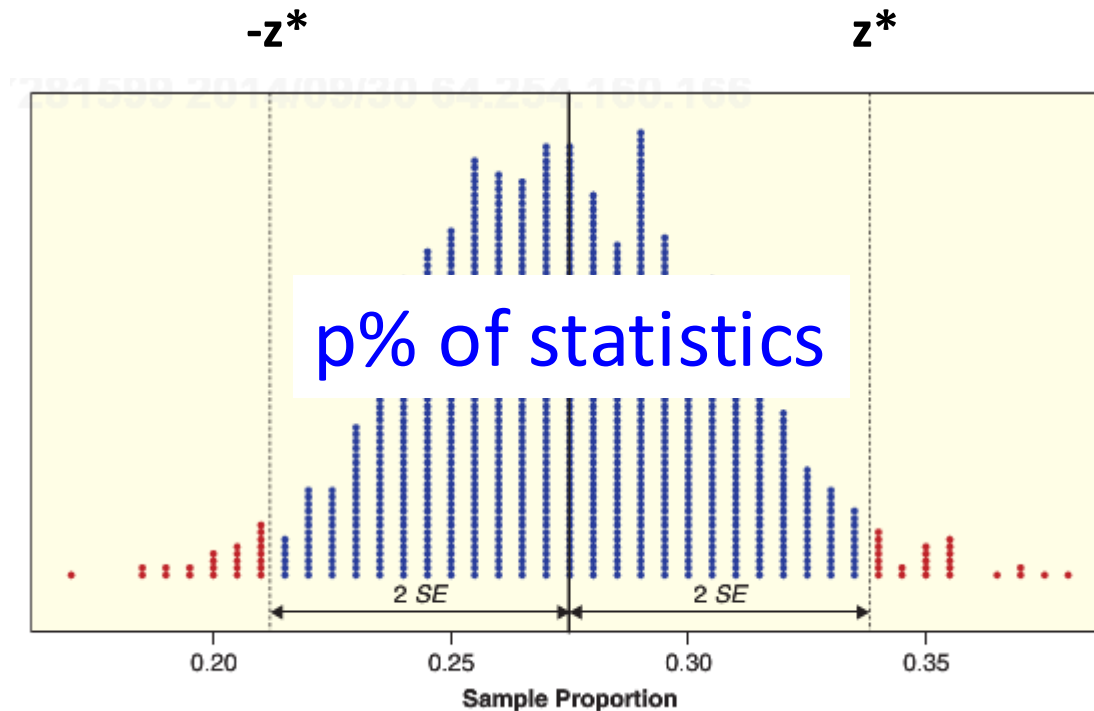
$$CI_{99.7} = \text{stat} \pm 3 \cdot SE$$

$$CI_{68} = \text{stat} \pm 1 \cdot SE$$

Confidence intervals for other confidence levels

Q₁₇: How could we get a confidence interval for the q^{th} confidence level?

A: We need to find the **critical value** z^* such that $p\%$ of our statistics are within $\pm z^* \cdot \text{SE}$ for a standard normal distribution



$$CI = \text{stat} \pm z^* \cdot \text{SE}$$

```
library(SDS1000)
```

```
cnorm(0.95)
```

```
[1] 1.96
```

Sampling distributions

Q₁₈: Could we calculate the SE by repeatedly sampling from a population to create sampling distribution, and then take the sd of this sampling distribution?

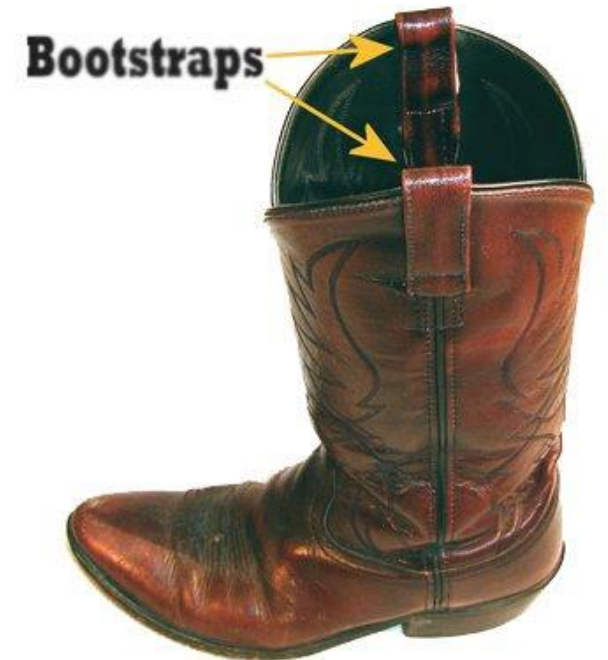


Sampling distributions

Q₁₉: If we can't calculate the sampling distribution, what else can we do?

- We can pick ourselves up from the bootstraps

1. Estimate SE with $\hat{SE}$ ***from a single sample of data***
2. Then use $\bar{x} \pm 2 \cdot \hat{SE}$ to get the 95% CI



Confident intervals

Q₂₀: Do you feel confident about confidence intervals?

- I hope so!

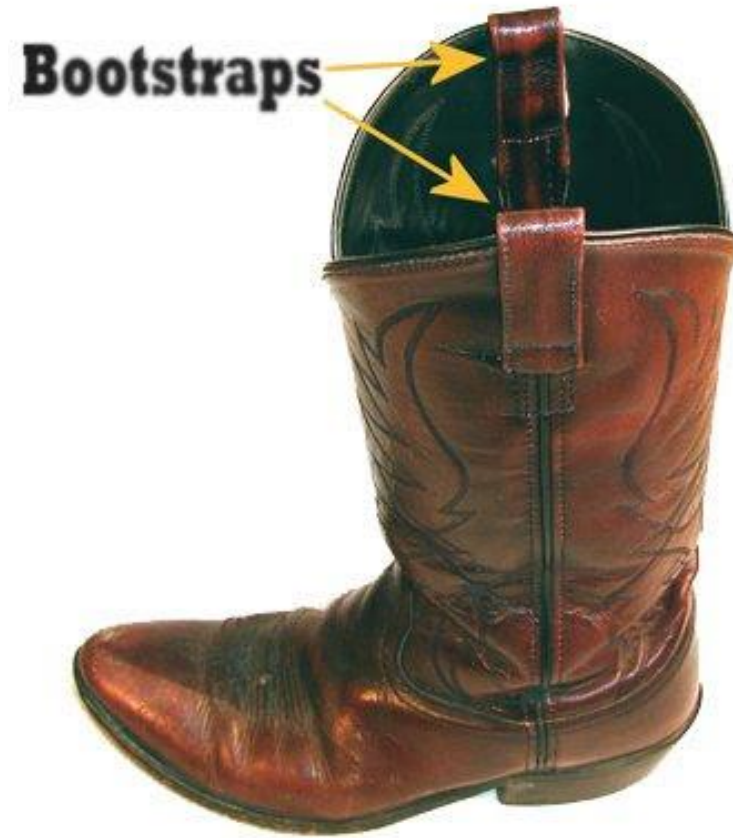
Q₂₁: If you don't feel confident about confidence intervals, how can you become more confident?

- Go to the practice sessions
- Review material
- Ask questions on Ed Discussion
- Come to office hours!



Let's quickly explore getting critical values in R!

The bootstrap



The bootstrap

The bootstrap is a method to estimate the standard error

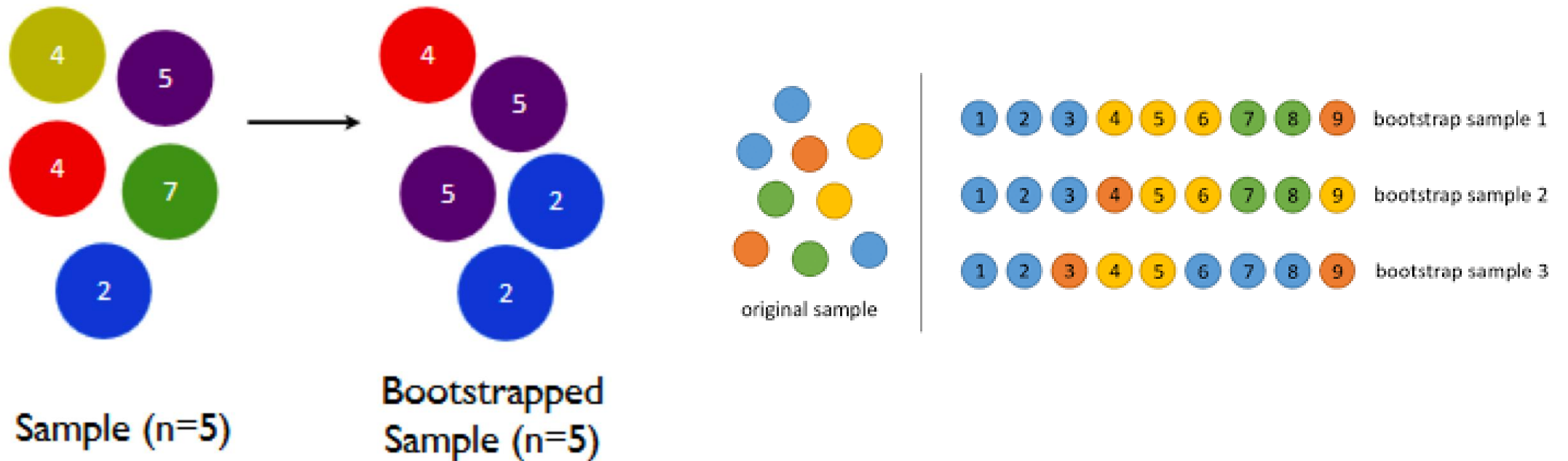
- $\hat{SE}$ is an estimate for SE
- We will use the symbol SE^* as the **bootstrap estimate** for SE (rather than $\hat{SE}$)

1. Estimate SE with SE^*
2. Then use $\bar{x} \pm 2 \cdot SE^*$ to get the 95% CI



Sampling with replacement

Sampling data **with replacement** means we can select some of the values more than once, and might not select other values at all



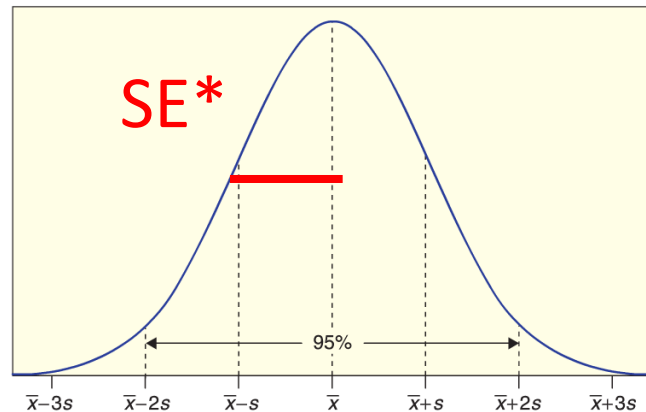
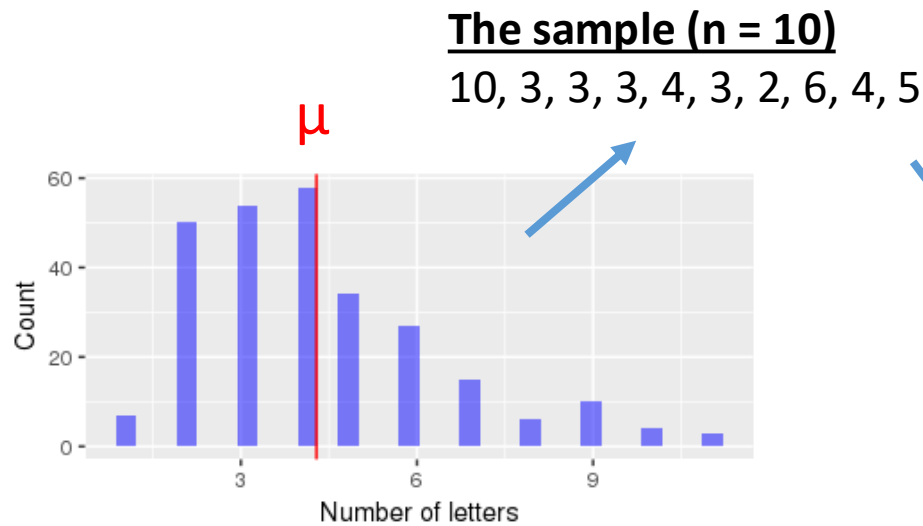
Plug-in principle

Suppose we get one sample of size n from a population

We pretend that this sample is the population (plug-in principle)

1. We then sample n points with replacement from **our sample**, and compute our statistic of interest
2. We repeat this process 1000's of times and get a *bootstrap* sample distribution
3. The standard deviation of this bootstrap distribution (SE* bootstrap) is a good approximate for standard error SE from the real sampling distribution

Gettysburg address word length bootstrap distribution



Bootstrap distribution!

3, 3, 3, 5, 3,
4, 5, 2, 2, 10

$$\bar{x}^* = 4$$

3, 3, 2, 3, 6,
4, 6, 5, 3, 6

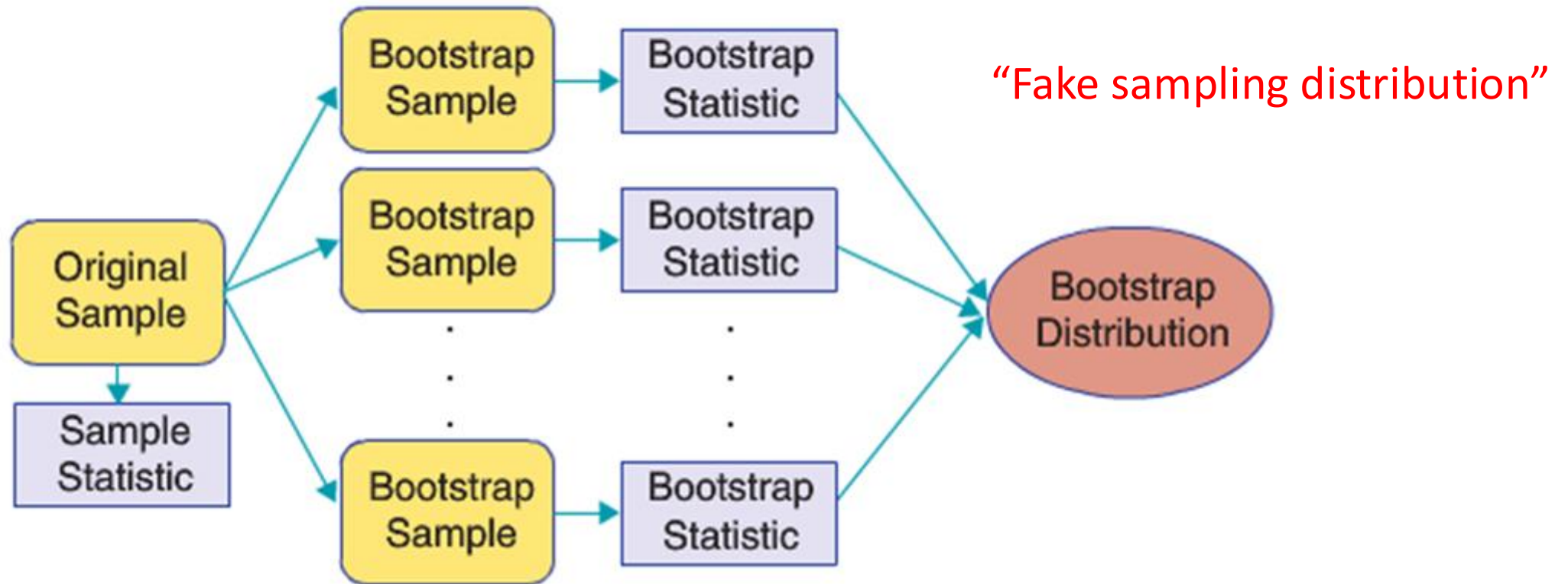
$$\bar{x}^* = 4.1$$

5, 3, 2, 3, 3,
3, 10, 3, 4, 3

$$\bar{x}^* = 3.9$$

Notice there is no 9's in the bootstrap samples

Bootstrap process



95% Confidence Intervals

When a bootstrap distribution for a sample statistic is approximately normal, we can estimate a 95% confidence interval using:

$$\textit{Statistic} \pm 2 \cdot SE^*$$

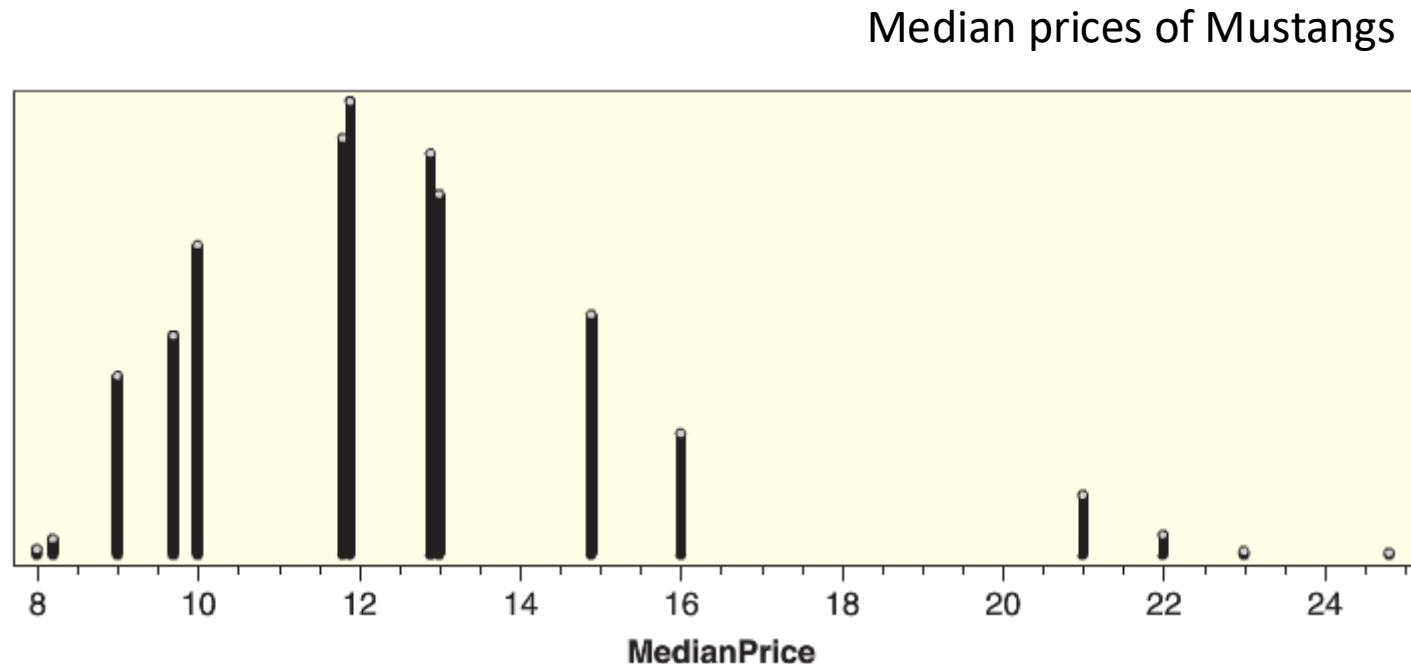
Where SE^* is the standard error estimated using the bootstrap

Findings CIs for many different parameters

The bootstrap method works for constructing confidence intervals for many different types of parameters!

Caution: the bootstrap does not always work

Always look at the bootstrap distribution, if it is poorly behaved (e.g., heavily skewed, has isolated clumps of values, etc.), you should not trust the intervals it produces.



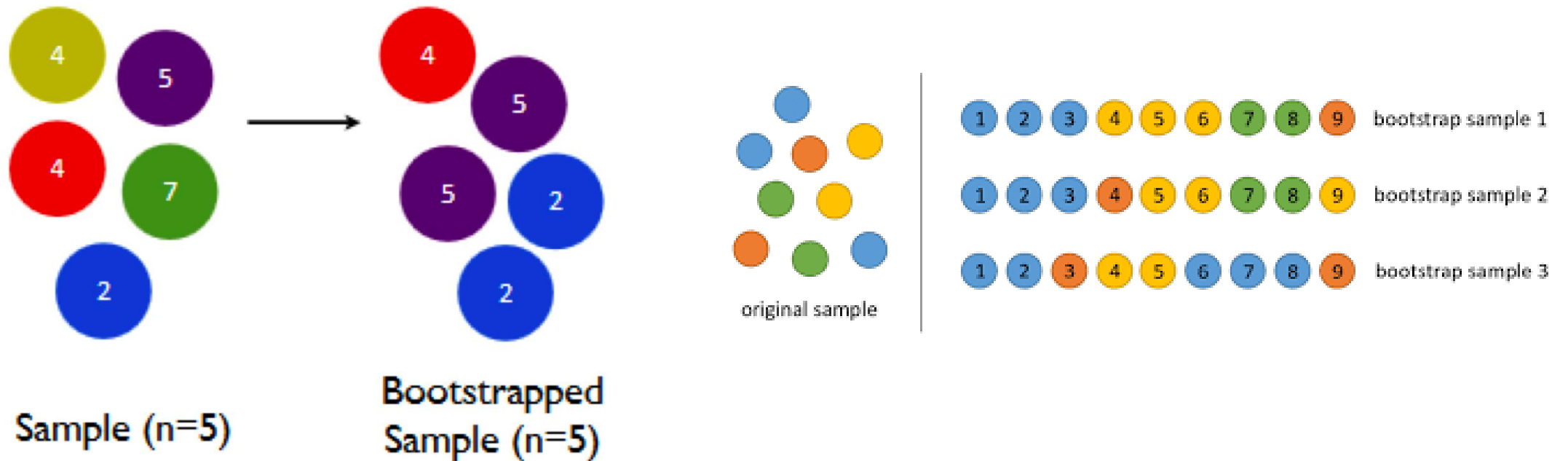
Calculating bootstrap confidence intervals in R

What are the steps needed to create a bootstrap SE?

1. Start with a sample
2. Repeat steps 10,000 times
 - a. Resample the points in the sample to get a bootstrap sample
 - b. Compute the statistic of interest on the bootstrap sample
3. Take the standard deviation of the bootstrap distribution to get SE^*

Sampling with replacement

Sampling data **with replacement** means we can select some of the values more than once, and might not select other values at all



Sampling with replacement from a vector

```
my_sample <- c(3, 1, 4, 1, 5, 9)
```

To get a sample of size n = 6 with replacement:

```
boot_sample <- sample(my_sample, 6, replace = TRUE)
```

Sampling distribution in R

```
my_sample <- c(21, 29, 25, 19, 24, 22, 25, 26, 25, 29)
```

```
bootstrap_dist <- do_it(10000) * {  
    curr_boot <- sample(my_sample , 10, replace = TRUE)  
    mean(curr_boot)  
}
```

```
SE_boot <- sd(bootstrap_dist)
```

Bootstrap confidence interval in R

```
obs_mean <- mean(my_sample)
```

```
CI_lower <- obs_mean - 2 * SE_boot
```

```
CI_upper <- obs_mean + 2 * SE_boot
```